CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

# Chapter 1: Rational and Irrational Numbers

Topic-wise short notes with solved examples

Rational numbers and their decimal forms, irrational numbers and surds, placing them on the number line, simplifying surds and rationalising denominators.

## 1 Rational numbers

- A **rational number** can be written as $\frac{p}{q}$, where $p$ and $q$ are integers and $q \neq 0$.
- Every integer, every fraction, every terminating decimal and every recurring decimal is rational.
- Between any two rational numbers there are infinitely many rational numbers. One of them is their mean: $\frac{a+b}{2}$.
- To insert $n$ rational numbers between $a$ and $b$, write both with a common denominator, multiply numerator and denominator by $n+1$, and pick the fractions in between.

**Example 1.1** Insert three rational numbers between $\frac{1}{3}$ and $\frac{1}{2}$.

SOLUTION

$\frac{1}{3} = \frac{2}{6} = \frac{8}{24}$ and $\frac{1}{2} = \frac{3}{6} = \frac{12}{24}$

$\frac{9}{24}, \frac{10}{24}, \frac{11}{24}$, that is $\frac{3}{8}, \frac{5}{12}, \frac{11}{24}$

## 2 Decimal form of a rational number

- The decimal form of a rational number either **terminates** or **recurs**.
- $\frac{p}{q}$ (in lowest terms) terminates exactly when $q$ has **no prime factor other than 2 and 5**.
- A recurring decimal is written with a bar: $0.\overline{3} = 0.333\ldots$, $0.1\overline{6} = 0.1666\ldots$
- **Recurring decimal to fraction**: let $x$ be the number, multiply by a power of 10 so that the recurring parts line up, and subtract.

**Example 2.1** Without dividing, say whether $\frac{7}{80}$ and $\frac{5}{12}$ terminate.

SOLUTION

$80 = 2^4 \times 5$: only 2 and 5. $12 = 2^2 \times 3$: contains 3.

$\frac{7}{80}$ terminates; $\frac{5}{12}$ recurs

**Example 2.2** Express $0.\overline{27}$ as a fraction.

SOLUTION

Let $x = 0.2727\ldots$ Then $100x = 27.2727\ldots$

$99x = 27$

$x = \frac{27}{99} = \frac{3}{11}$

**Example 2.3** Express $0.2\overline{3}$ as a fraction.

SOLUTION

$10x = 2.333\ldots$ and $100x = 23.333\ldots$, so $90x = 21$

$\frac{7}{30}$

## 3 Irrational numbers

- A number that **cannot** be written as $\frac{p}{q}$ is **irrational**. Its decimal form neither terminates nor recurs.
- Examples: $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, $\sqrt[3]{4}$, $\pi$, $0.101001000100001\ldots$
- $\sqrt{n}$ is irrational whenever $n$ is a positive integer that is not a perfect square.
- Rational and irrational numbers together form the **real numbers**. Every real number has a point on the number line.
- **Proof that $\sqrt{2}$ is irrational**: suppose $\sqrt{2} = \frac{p}{q}$ in lowest terms. Then $p^2 = 2q^2$, so $p$ is even, say $p = 2k$. Then $q^2 = 2k^2$, so $q$ is even too. This contradicts "lowest terms".

| Operation | Result |
|---|---|
| rational $\pm$ irrational | irrational |
| non-zero rational $\times$ irrational | irrational |
| irrational $\pm$ irrational | may be rational or irrational |
| irrational $\times$ irrational | may be rational or irrational |

**Example 3.1** Show that $3 + \sqrt{5}$ is irrational.

SOLUTION

Suppose $3 + \sqrt{5} = r$, a rational number. Then $\sqrt{5} = r - 3$, which is rational.

**This contradicts the fact that $\sqrt{5}$ is irrational, so $3 + \sqrt{5}$ is irrational**

**Example 3.2** Write two irrational numbers between 2 and 3.

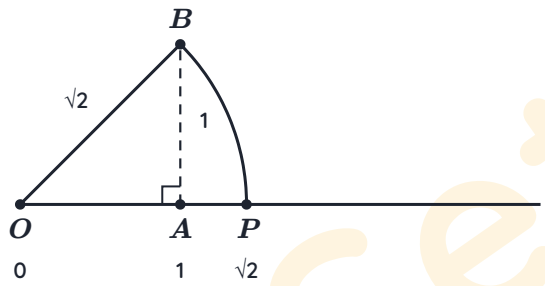
SOLUTION

Any $\sqrt{n}$ with $4 < n < 9$ and $n$ not a perfect square will do.

$\sqrt{5}$ and $\sqrt{7}$

## 4 Surds on the number line

- $\sqrt{2}$: draw a right triangle with both shorter sides 1 unit on the number line from 0. Its hypotenuse is $\sqrt{2}$. With centre 0 and this radius, cut the number line.
- $\sqrt{3}$: use shorter sides $\sqrt{2}$ and 1. $\sqrt{5}$: use shorter sides 2 and 1.




OA = 1 and AB = 1, so OB = √2. The arc with centre O marks P at √2.

## 5 Surds

- If $\sqrt[n]{a}$ is irrational and $a$ is a positive rational number, then $\sqrt[n]{a}$ is a **surd** of order $n$. $\sqrt{3}$ is a surd; $\sqrt{9}$ and $\pi$ are not.
- $\sqrt{ab} = \sqrt{a} \times \sqrt{b}$ and $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$ for positive $a, b$.
- **Like surds** have the same irrational part and can be added: $3\sqrt{2} + 5\sqrt{2} = 8\sqrt{2}$.
- **Simplest form**: take out the largest perfect-square factor. $\sqrt{72} = \sqrt{36 \times 2} = 6\sqrt{2}$.

**Example 5.1** Simplify $\sqrt{50} + \sqrt{18} - \sqrt{8}$.

SOLUTION

$5\sqrt{2} + 3\sqrt{2} - 2\sqrt{2}$

$6\sqrt{2}$

**Example 5.2** Expand $(2\sqrt{3} + \sqrt{5})(2\sqrt{3} - \sqrt{5})$.

SOLUTION

$(2\sqrt{3})^2 - (\sqrt{5})^2 = 12 - 5$

**7**

**Example 5.3** Which is greater, $3\sqrt{2}$ or $2\sqrt{5}$?

SOLUTION

$3\sqrt{2} = \sqrt{18}$ and $2\sqrt{5} = \sqrt{20}$

$2\sqrt{5}$

## 6 Rationalisation

- **Rationalising** a denominator means removing the surd from it by multiplying numerator and denominator by a suitable **rationalising factor**.

| Denominator | Rationalising factor |
|---|---|
| $\sqrt{a}$ | $\sqrt{a}$ |
| $a + \sqrt{b}$ | $a - \sqrt{b}$ |
| $\sqrt{a} + \sqrt{b}$ | $\sqrt{a} - \sqrt{b}$ |
| $\sqrt{a} - \sqrt{b}$ | $\sqrt{a} + \sqrt{b}$ |

- $a + \sqrt{b}$ and $a - \sqrt{b}$ are **conjugates**. Their product $a^2 - b$ is rational.

**Example 6.1** Rationalise the denominator of $\frac{5}{\sqrt{3}}$.

SOLUTION

$\frac{5\sqrt{3}}{3}$

**Example 6.2** Rationalise the denominator of $\frac{4}{3 - \sqrt{5}}$.

SOLUTION

$\frac{4(3 + \sqrt{5})}{9 - 5}$

$3 + \sqrt{5}$

**Example 6.3** If $\frac{\sqrt{3} + 1}{\sqrt{3} - 1} = a + b\sqrt{3}$, find $a$ and $b$.

SOLUTION

$\frac{(\sqrt{3} + 1)^2}{3 - 1} = \frac{4 + 2\sqrt{3}}{2} = 2 + \sqrt{3}$

$a = 2, b = 1$

**Example 6.4** If $x = 2 + \sqrt{3}$, find $x + \frac{1}{x}$ and $x^2 + \frac{1}{x^2}$.

SOLUTION

$\frac{1}{x} = \frac{2 - \sqrt{3}}{4 - 3} = 2 - \sqrt{3}$, so $x + \frac{1}{x} = 4$.

$x^2 + \frac{1}{x^2} = 4^2 - 2$

**4 and 14**

## 7 Common mistakes

- Writing $\sqrt{a + b} = \sqrt{a} + \sqrt{b}$. This is false: $\sqrt{9 + 16} = 5$, not 7.
- Calling $\frac{22}{7}$ irrational. It is rational; it is only an approximation for $\pi$.
- Adding unlike surds: $\sqrt{2} + \sqrt{3}$ is not $\sqrt{5}$.
- Multiplying only the denominator by the rationalising factor.
- Using $a + \sqrt{b}$ again instead of its conjugate.
- Forgetting to reduce $\frac{p}{q}$ to lowest terms before testing the denominator for 2s and 5s.

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